SYSTEM-BASED INTERACTION DESIGN

PRESENTED BY

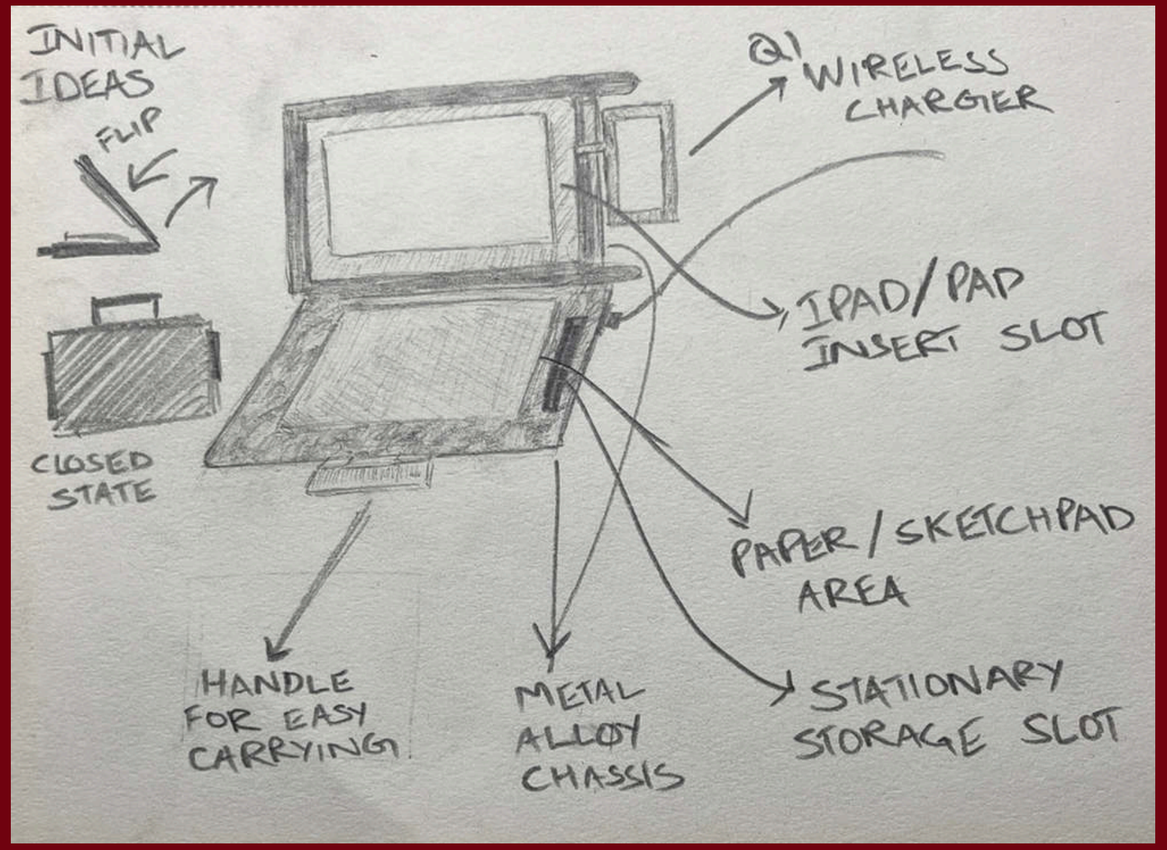
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BDES(IXD)

MOBILE WORK STATION

PROBLEM

1. A User-Centered Design exploration that produced an integrated mobile creative workstation — a portable "base of operations" for architects, designers, and illustrators

2. Central design intent: bridge the gap between digital precision (iPad, stylus, Maya, Procreate) and the tactile immediacy of analog sketching, packaging both into a single deployable platform



The "Footprint" Problem — users in third spaces (cafés, libraries, transit) struggle to fit an iPad, physical sketchbook, and pencil case onto the small surfaces available

Framed around the central tension identified as the “Digital-Analog Divide”- a lot of creative professionals tend to hover between two distinct working modes

DISCOVER

Cable Anxiety — mobile workflows depend on a fragile choreography of power banks, tangled cables, and adapters just to keep the iPad and peripherals alive across a session

Contextual Environment — urban creators work in cramped, high-traffic settings where equipment safety and rapid setup are mechanical necessities, not luxuries

DEFINE

1

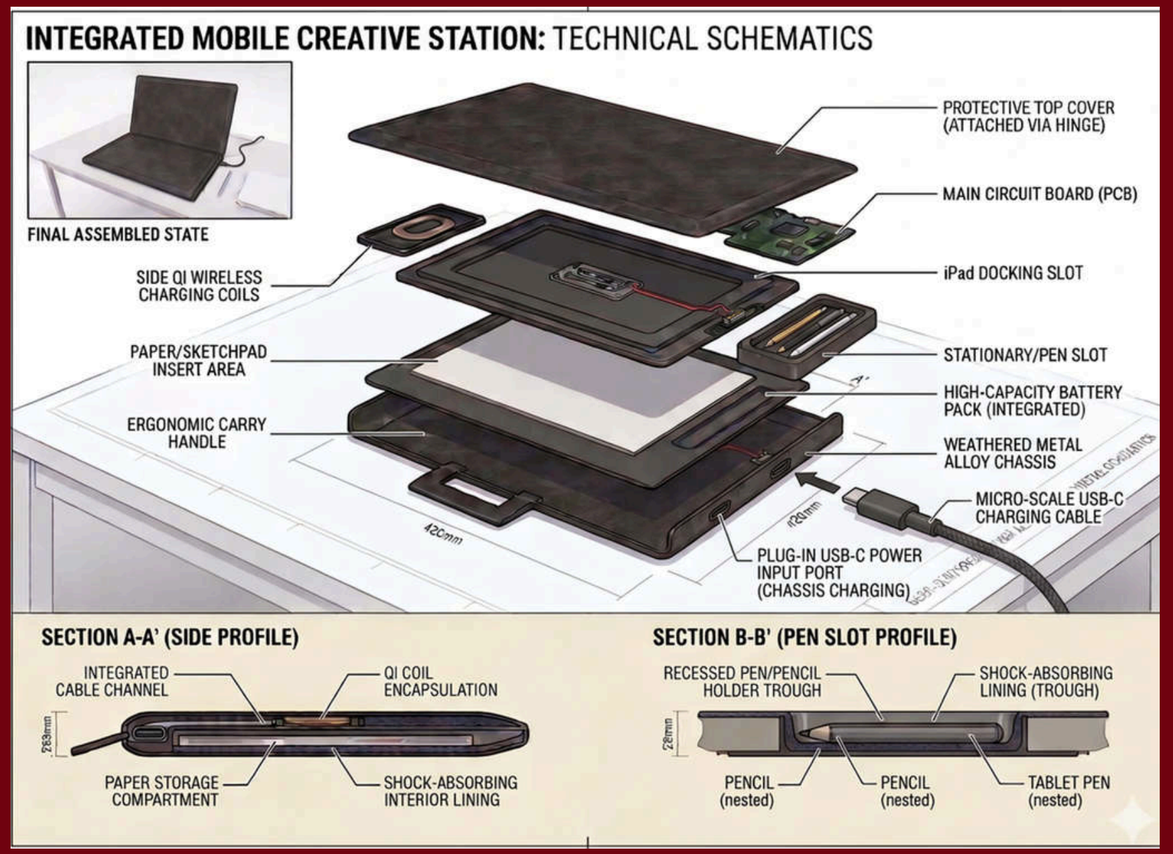
PROBLEM STATEMENT — CREATIVE WORKFLOWS FORCE USERS INTO CHOOSING BETWEEN DIGITAL TABLETS AND MANUAL SKETCHING, ULTIMATELY BEING FACED BY POOR POWER MANAGEMENT AND THE BURDEN OF CARRYING MULTIPLE DISJOINTED TOOLS

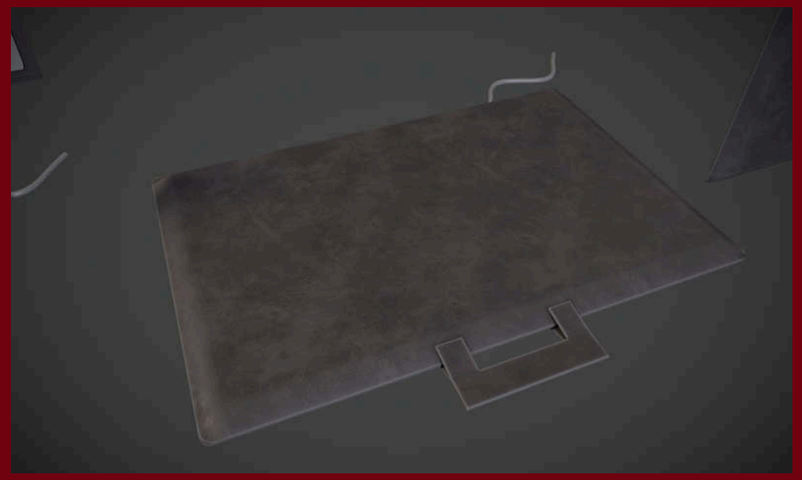
2

DESIGN CHALLENGE — CREATE A UNIFIED, PORTABLE PLATFORM THAT MINIMISES SETUP FRICTION, MAXIMISES COMFORT, AND LETS USERS TRANSITION BETWEEN 3D MODELING AND PHYSICAL SKETCHING WITHOUT ENLARGING THEIR WORKSPACE FOOTPRINT

DELIVER




INTEGRATED MOBILE CREATIVE STATION: TECHNICAL SCHEMATICS
FINAL ASSEMBLED STATE
PROTECTIVE TOP COVER (ATTACHED VIA HINGE)
MAIN CIRCUIT BOARD (PCB)
iPad DOCKING SLOT
SIDE QI WIRELESS CHARGING COILS
PAPER/SKETCHPAD INSERT AREA
STATIONARY/PEN SLOT
HIGH-CAPACITY BATTERY PACK (INTEGRATED)
ERGONOMIC CARRY HANDLE
WEATHERED METAL ALLOY CHASSIS
MICRO-SCALE USB-C CHARGING CABLE
PLUG-IN USB-C POWER INPUT PORT (CHASSIS CHARGING)
420mm
SECTION A-A' (SIDE PROFILE)
INTEGRATED CABLE CHANNEL
QI COIL ENCAPSULATION
PAPER STORAGE COMPARTMENT
SHOCK-ABSORBING INTERIOR LINING
SECTION B-B' (PEN SLOT PROFILE)
RECESSED PEN/PENCIL HOLDER TROUGH
SHOCK-ABSORBING LINING (TROUGH)
PENCIL (nested)
PENCIL (nested)
TABLET PEN (nested)







ATTENDANCE PORTAL

PROBLEM

1. Facial recognition is unreliable in practice — the in-class scanner frequently lags, fails to detect faces, or misidentifies students, leaving attendance unrecorded despite the student being physically present

2. No mechanism to prevent proxy in the manual flow — when professors mark attendance manually, they often go by a quick visual sweep, which is vulnerable to the exact proxy attendance the facial recognition was meant to eliminate

3. Manual fallback wastes class time — when the scanner fails, the professor has to open their own app and mark each student one by one, consuming 5–10 minutes of teaching time per session

4. The core issue is over-reliance on a single point of failure — the entire process depends on the camera working perfectly, and when it doesn't, there is no graceful, fast,alternative

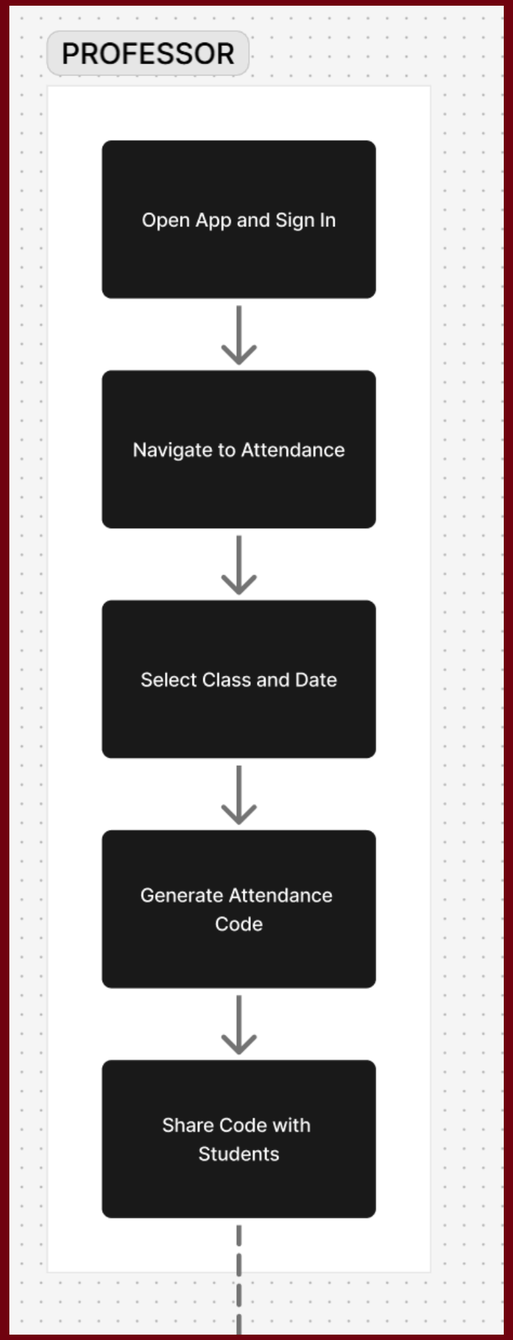
USER JOURNEY

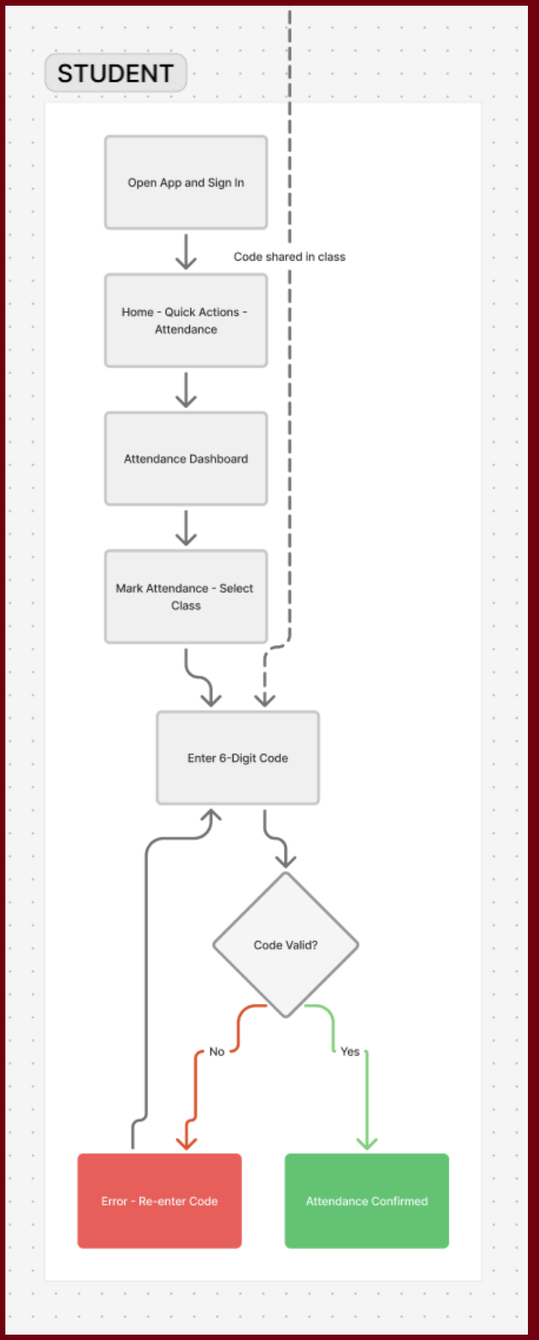
STUDENT

- Open App — login via Student ID + PIN
- Session Notification — push alert when faculty opens a session
- Enter Code — type the rotating 6-digit faculty code
- Geo Check — app silently verifies student is inside classroom radius
- Confirmed — immediate success screen with timestamp; record locked

PROFESSOR

- Login → Select Class → Start Session — system generates rotating code + activates geofence
- Share Code — displayed on projector or announced verbally; rotates every 15 sec
- Monitor Live Dashboard — real-time Total / Present / Absent counts with per-student timestamps
- Close Session — one-tap finalise; records become immutable





ANTI-CHEATING

Layer 1 — Rotating Code (15-sec reset) defeats screenshot sharing; by the time a screenshot reaches an absent student, the code has changed

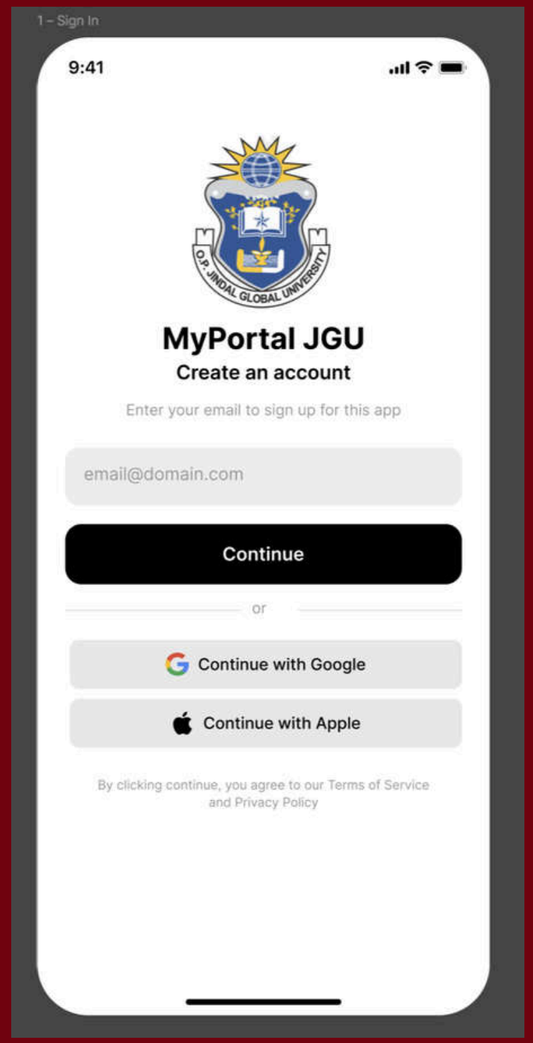
Layer 2 — Geofencing (GPS) → defeats remote code-sharing even if a friend sends the code, the absent student's GPS fails the proximity check

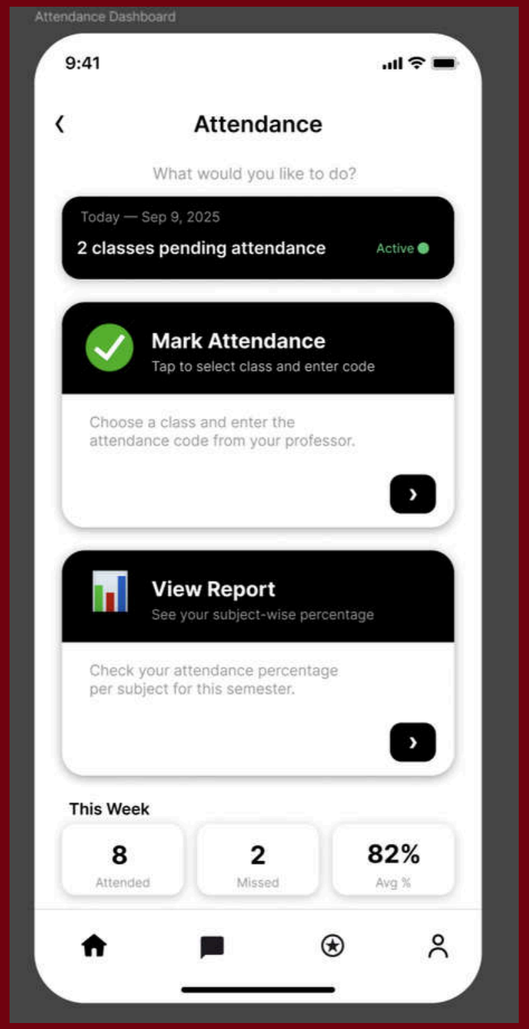
Layer 3 — Device Binding → defeats phone-handover proxy- account is locked to one device ID, second-device login is flagged and rejected

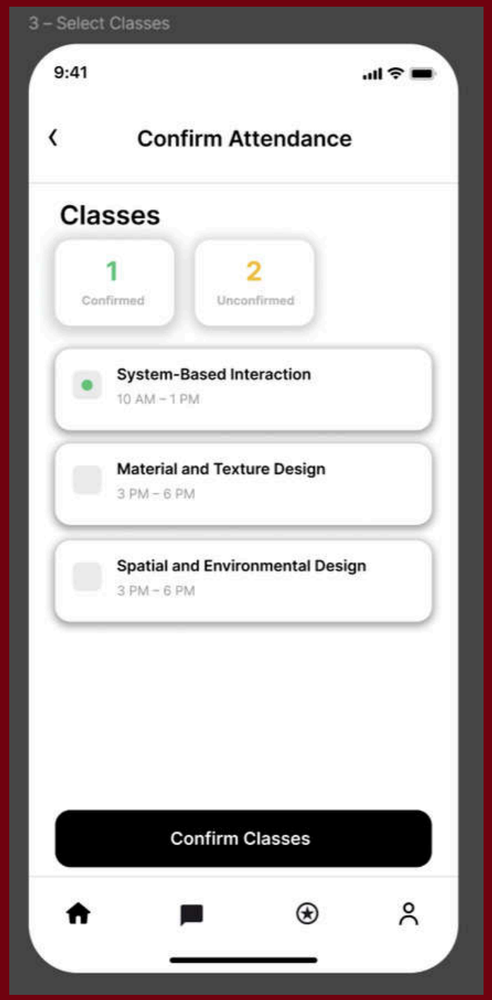
Layer 4 — Session Time Window defeats backdating; hard start/end times are server-enforced

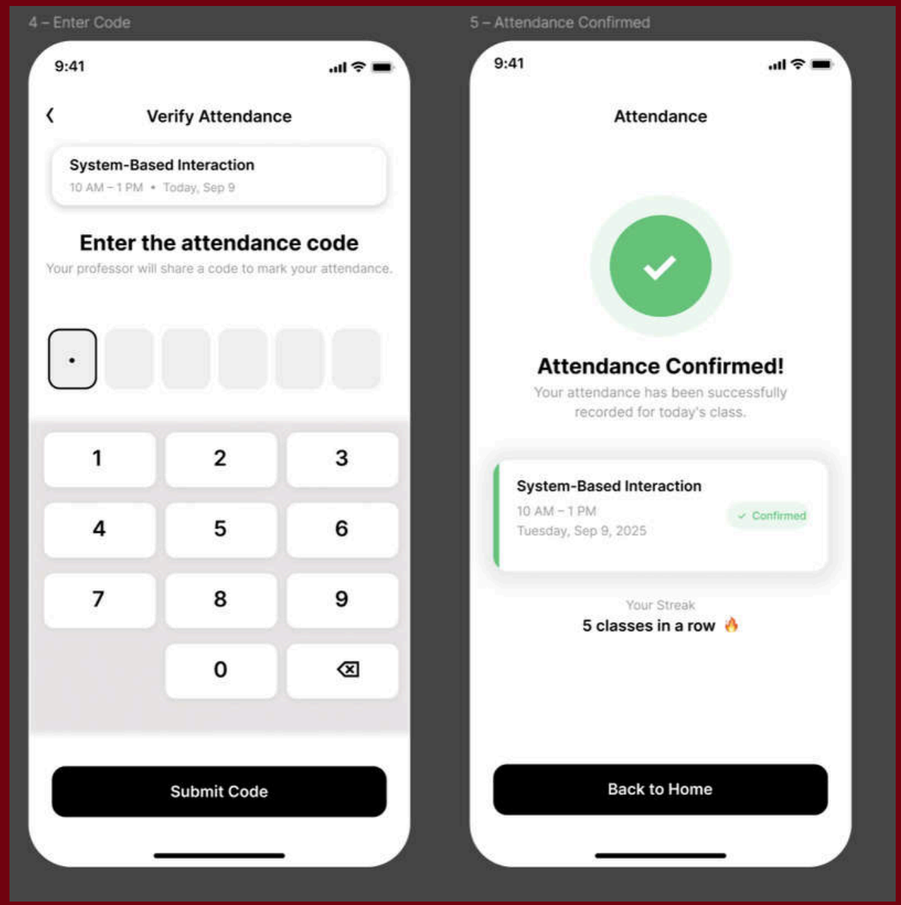
Layer 5 — Anomaly Detection → catches edge cases like GPS spoofing and flags for admin review, preserves evidence

UI DESIGN


1 – Sign In
9:41
MyPortal JGU
Create an account
Enter your email to sign up for this app
email@domain.com
Continue
or
Continue with Google
Continue with Apple
By clicking continue, you agree to our Terms of Service and Privacy Policy


Attendance Dashboard
9:41
Attendance
What would you like to do?
Today — Sep 9, 2025
2 classes pending attendance
Active
Mark Attendance
Tap to select class and enter code
Choose a class and enter the attendance code from your professor.
View Report
See your subject-wise percentage
Check your attendance percentage per subject for this semester.
This Week
8
Attended
2
Missed
82%
Avg %

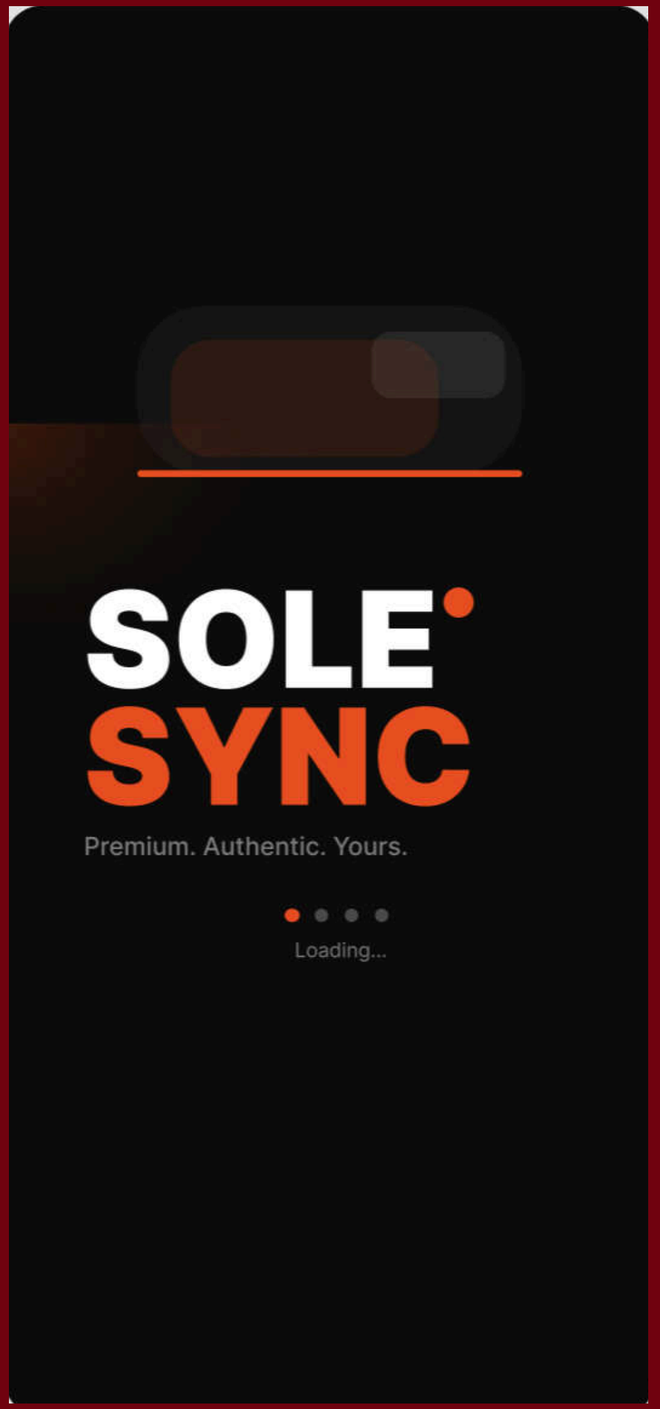

3 – Select Classes
9:41
Confirm Attendance
Classes
1
Confirmed
2
Unconfirmed
System-Based Interaction
10 AM – 1 PM
Material and Texture Design
3 PM – 6 PM
Spatial and Environmental Design
3 PM – 6 PM
Confirm Classes


4 – Enter Code
9:41
Verify Attendance
System-Based Interaction
10 AM – 1 PM • Today, Sep 9
Enter the attendance code
Your professor will share a code to mark your attendance.
1
2
3
4
5
6
7
8
9
0
Submit Code
5 – Attendance Confirmed
9:41
Attendance
Attendance Confirmed!
Your attendance has been successfully recorded for today's class.
System-Based Interaction
10 AM – 1 PM
Tuesday, Sep 9, 2025
Confirmed
Your Streak
5 classes in a row
Back to Home

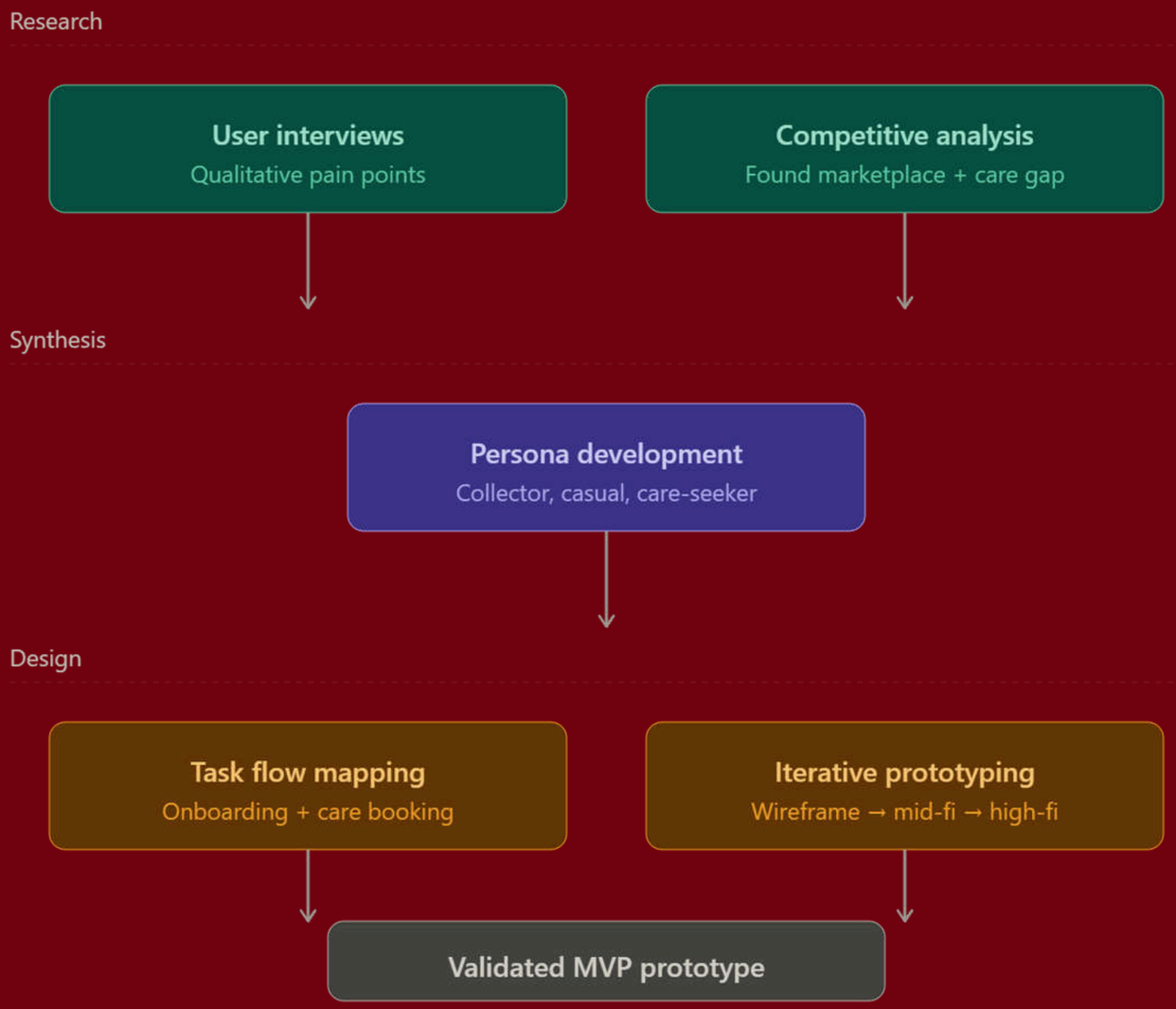
SOLE-SYNC

CONCEPT

SoleSync is an all-in-one sneaker platform that unifies the marketplace experience with an integrated shoe maintenance and care service in a single mobile app. Designed for both serious sneakerheads and casual buyers, it owns the full sneaker lifecycle — discovery, purchase, maintenance, and re-engagement — within one continuous ecosystem. Its core differentiator is a built-in care system that turns shoe maintenance from an everyday problem into a designated service in under a minute, transforming what would otherwise be a standard shopping app into a complete sneaker lifecycle platform.



RESEARCH AND DESIGN

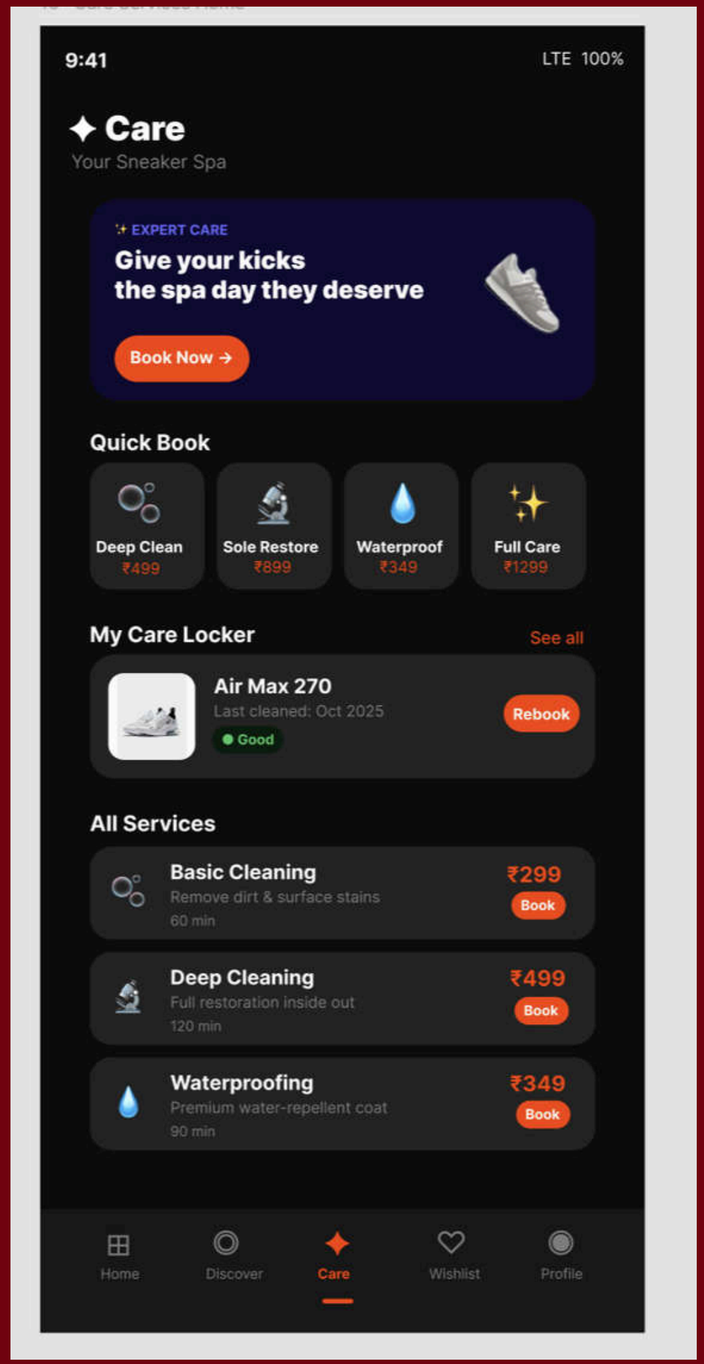


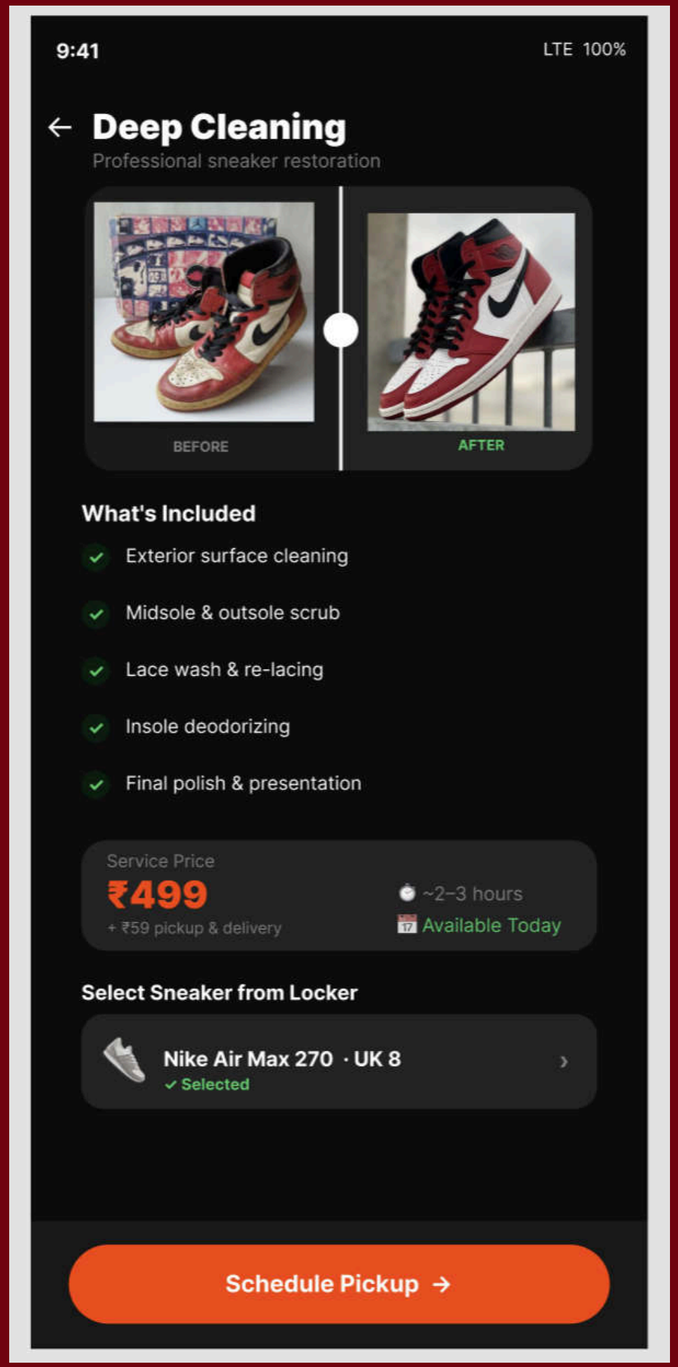
- **User Interviews — qualitative conversations with sneakerheads and casual buyers to surface real pain points**
- **Competitive Analysis — teardown of existing platforms that revealed the unfilled marketplace + maintenance gap**
- **Persona Development — archetypes from hardcore collector to casual buyer to maintenance-seeker**
- **Task Flow Mapping — detailed mapping of the two highest-stakes flows (onboarding and maintenance booking)**
- **Iterative Prototyping — wireframes with user feedback integrated between each stage**

USP

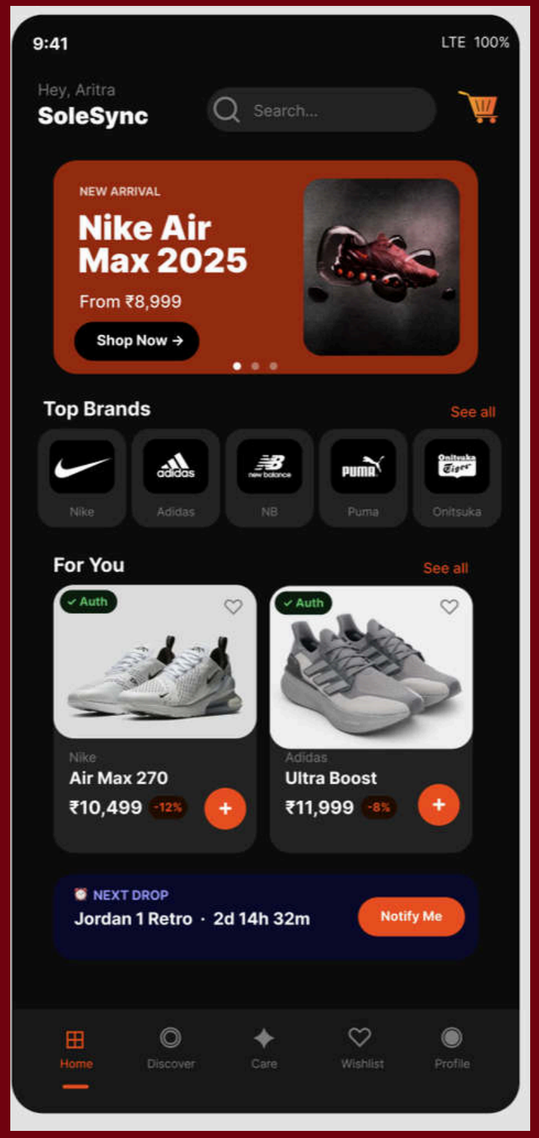
FEATURES

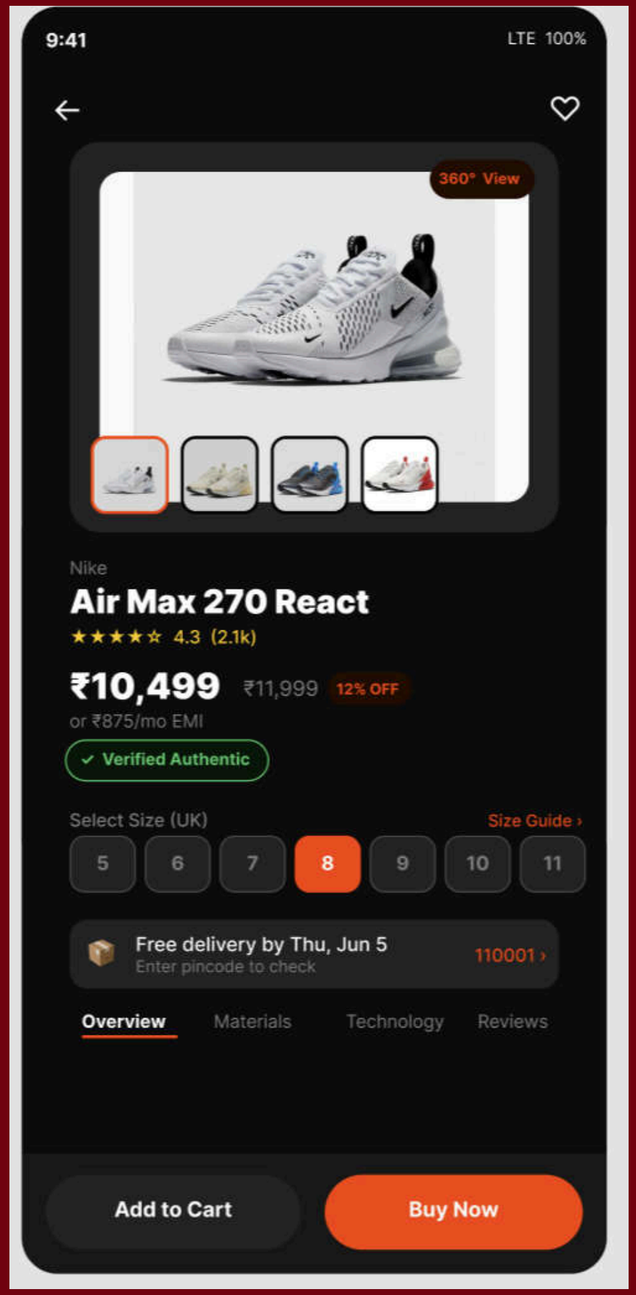
- First sneaker app with a fully integrated care service — competitors stop at the sale; SoleSync owns what comes after
- AI shoe scanner — point-and-scan damage assessment with instant, transparent pricing
- End-to-end service inside the app — scan to pickup to delivery, no offline handoff
- Multiple service tiers — cleaning, restoration, repair, customisation — each clearly priced

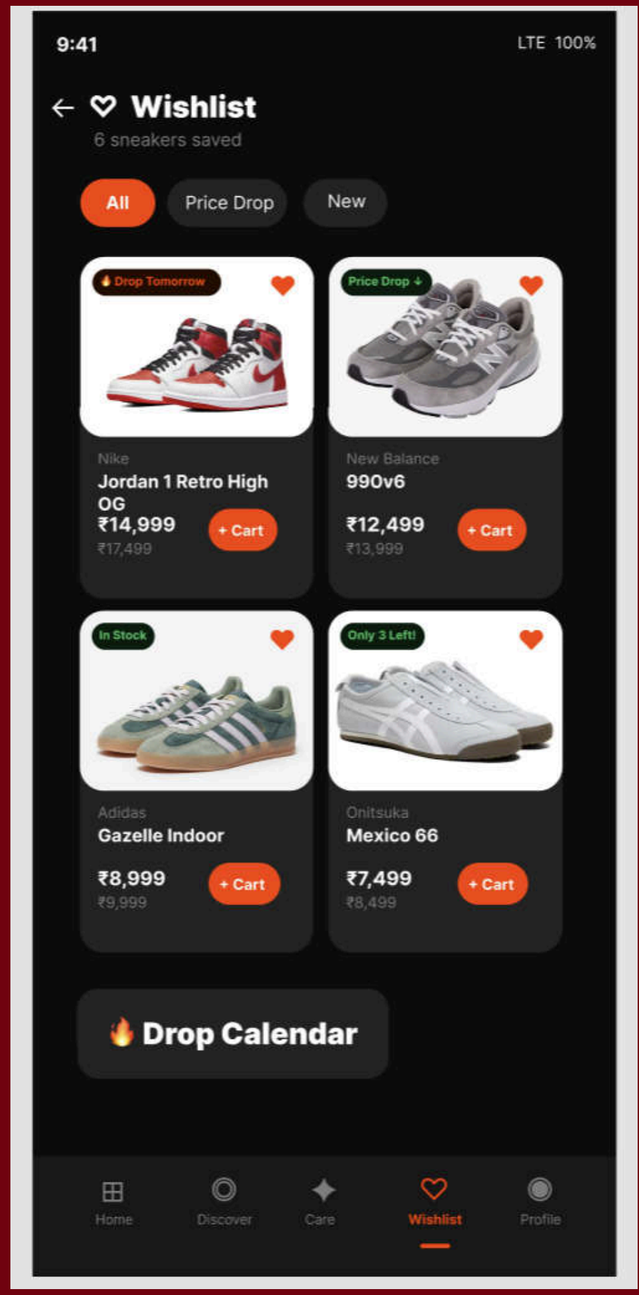


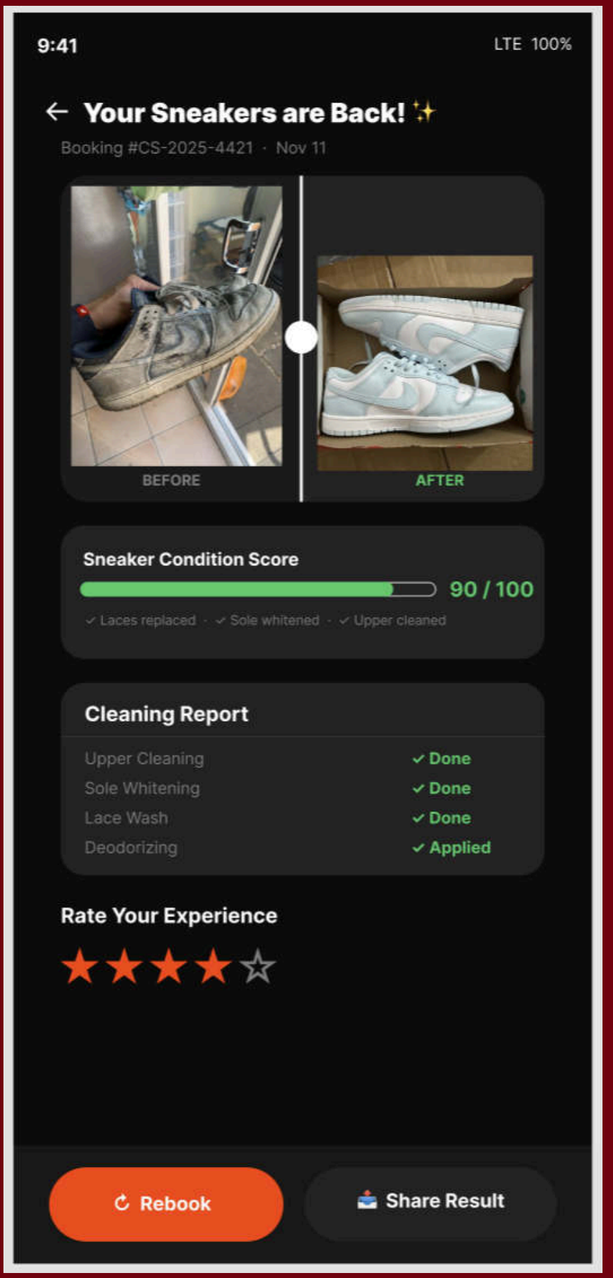


UI DESIGN


9:41
LTE 100%
Hey, Aritra
SoleSync
Search...
NEW ARRIVAL
Nike Air Max 2025
From ₹8,999
Shop Now →
Top Brands
See all
Nike
Adidas
NB
Puma
Onitsuka
For You
See all
Auth
Nike
Air Max 270
₹10,499 -12%
Auth
Adidas
Ultra Boost
₹11,999 -8%
NEXT DROP
Jordan 1 Retro · 2d 14h 32m
Notify Me
Home
Discover
Care
Wishlist
Profile


9:41
LTE 100%
360° View
Nike
Air Max 270 React
★★★★☆ 4.3 (2.1k)
₹10,499 ₹11,999 12% OFF
or ₹875/mo EMI
Verified Authentic
Select Size (UK)
Size Guide ›
5
6
7
8
9
10
11
Free delivery by Thu, Jun 5
Enter pincode to check
110001 ›
Overview
Materials
Technology
Reviews
Add to Cart
Buy Now


9:41
LTE 100%
Wishlist
6 sneakers saved
All
Price Drop
New
Drop Tomorrow
Nike
Jordan 1 Retro High OG
₹14,999
₹17,499
+ Cart
Price Drop ↓
New Balance
990v6
₹12,499
₹13,999
+ Cart
In Stock
Adidas
Gazelle Indoor
₹8,999
₹9,999
+ Cart
Only 3 Left!
Onitsuka
Mexico 66
₹7,499
₹8,499
+ Cart
Drop Calendar
Home
Discover
Care
Wishlist
Profile


9:41
LTE 100%
Your Sneakers are Back! ✨
Booking #CS-2025-4421 · Nov 11
BEFORE
AFTER
Sneaker Condition Score
90 / 100
✓ Laces replaced · ✓ Sole whitened · ✓ Upper cleaned
Cleaning Report
Upper Cleaning ✓ Done
Sole Whitening ✓ Done
Lace Wash ✓ Done
Deodorizing ✓ Applied
Rate Your Experience
Rebook
Share Result

THANK YOU